Control System Lab Final Exam

Exp. From 5 to 8

تم حل هذا الملف من قبل الطالب : ليث عمرو

هذه الحلول هي جهد بشري قابل للخطأ،
فإن أحسنت فمن الله وإن أخطأت فمنّي ومن الشيطان.

هذا العمل مقدّم ثوابه لروح والدي (قاسم مفيد عمرو)
فمن وجد فيه فائدة فليدعُ له بالرحمة والمغفرة.

تاريخ الحل : الفصل الثاني 2024-2025

(لا تنسى ترجع تحل ملف الميد لأنه ممكن يجي عليه)

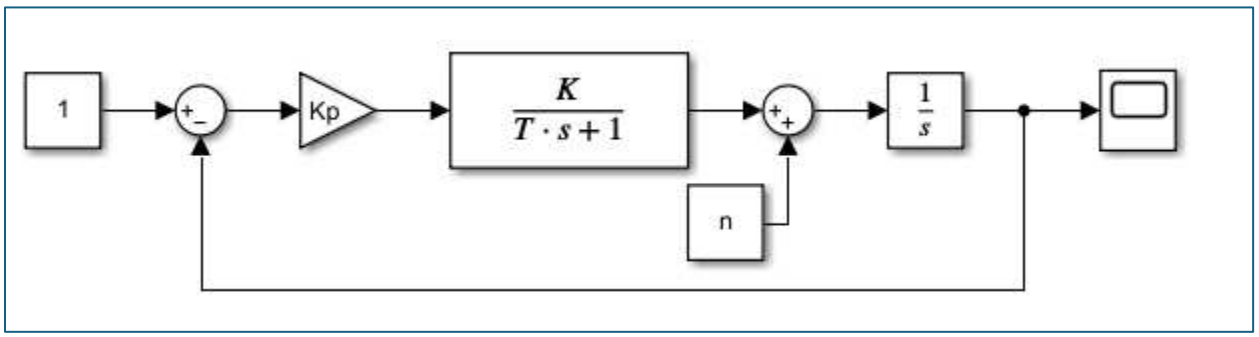
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Q1) In second order system, discuss the following:

Varying Kp in P Controller:



Solution:

When n = 0 :

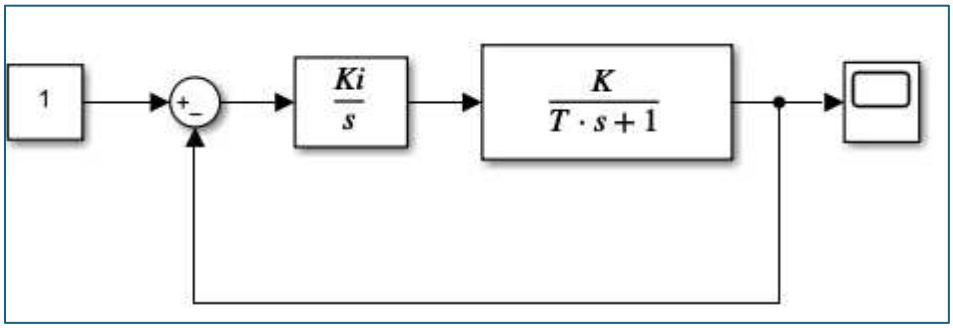
Kp (increase) → Oscillation (increase) → Mp (increase) → tr, td, tp (decrease)

When n = 1 :

Kp (increase) → Oscillation (increase) → Ess (decrease)

Q2) In first order system, discuss the following:

Varying Ki in I Controller:



Solution:

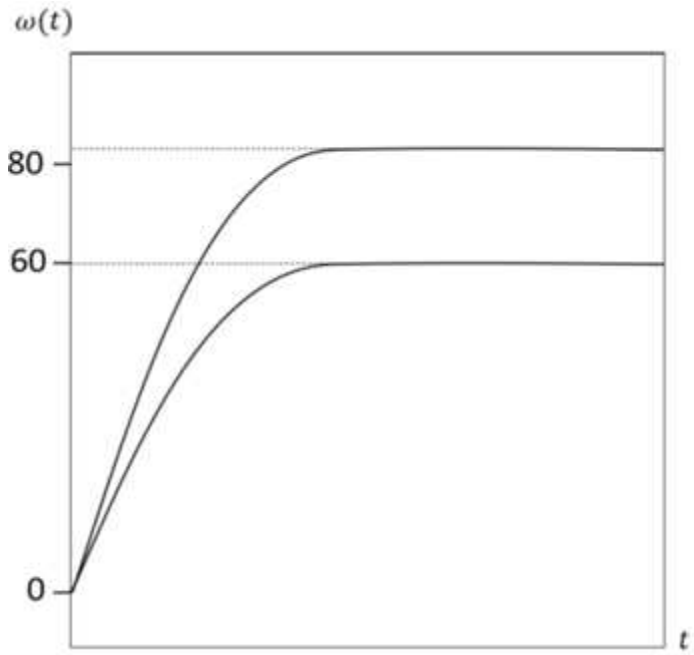
Ki (increase) → Oscillation (increase) → Mp (increase) → tr, td, tp (decrease)

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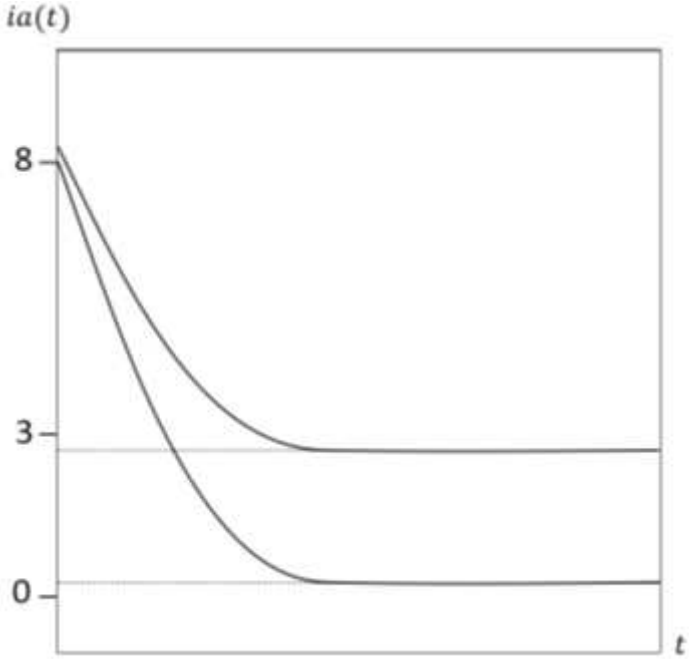
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Q3) The following curves are the output of dc motor:



Speed VS Time



Current VS Time

a) Which plot represents the case of no load in each curve ?

b) Explain the effect of load on i_a and ω ?

c) Sketch i_a and θ in case of position closed loop system for the above case ?

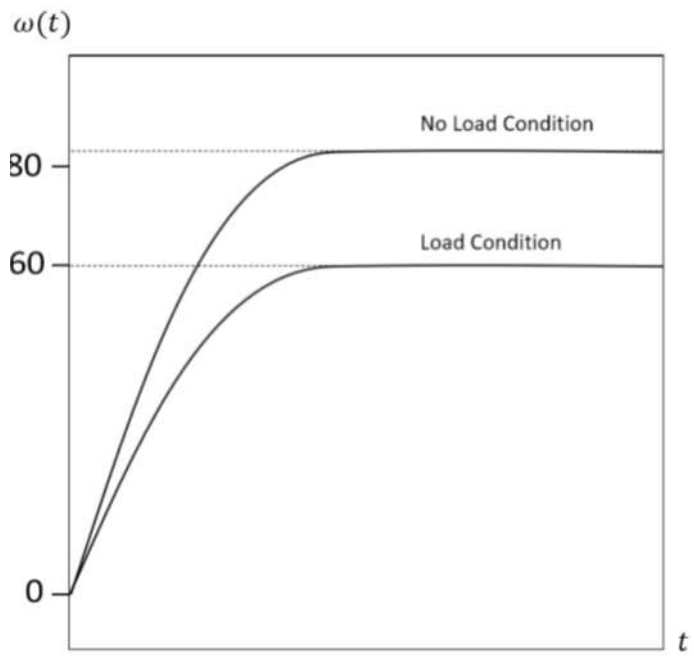
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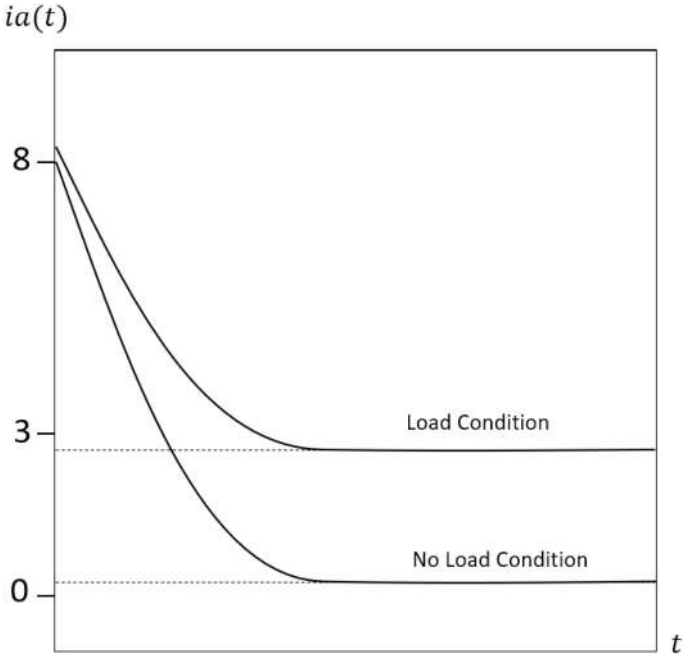
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Solution:

a)



Speed VS Time



Current VS Time

b) When no load:

ω increase

i_a decrease

When load:

ω decrease

i_a increase

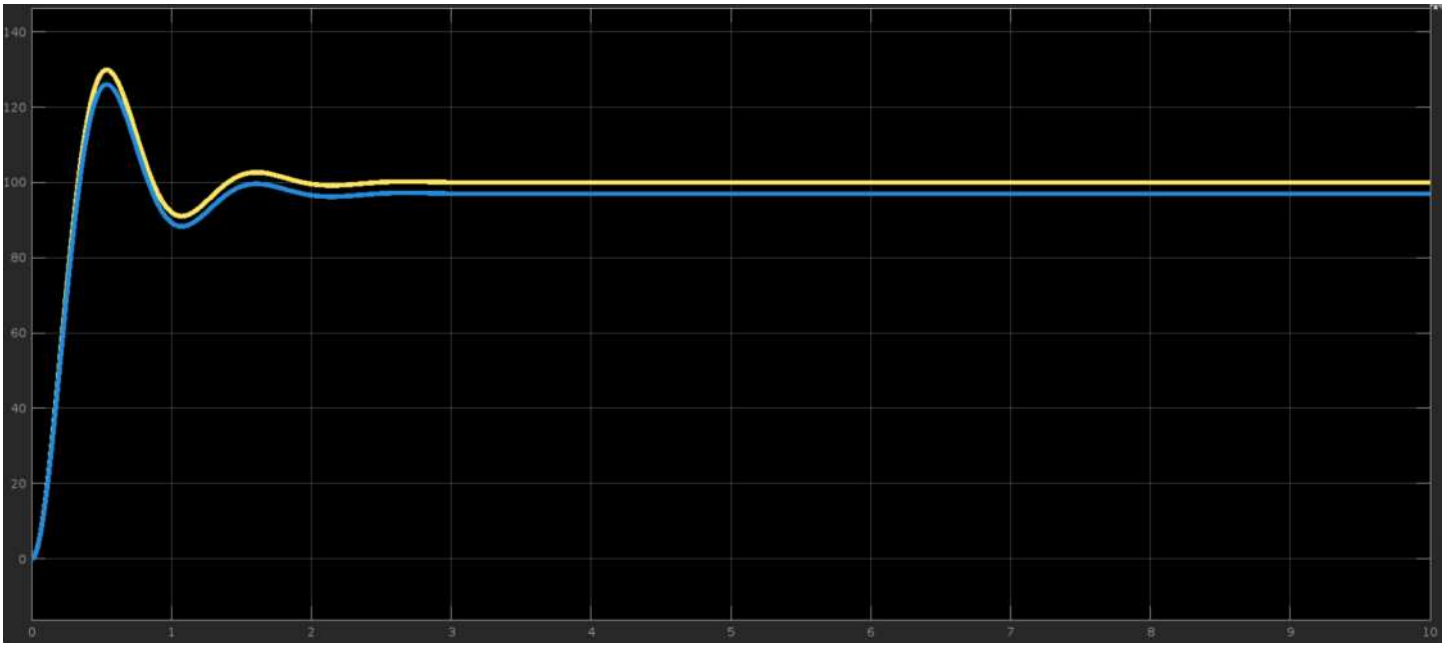
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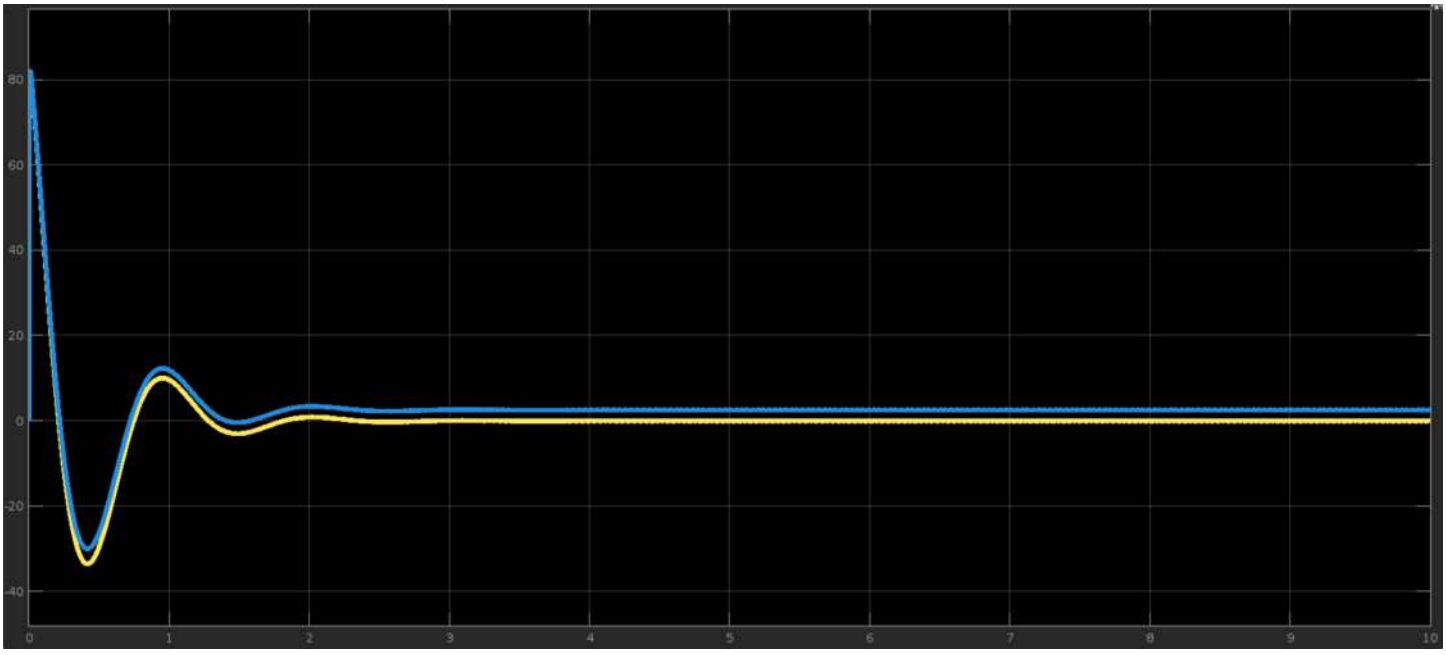
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c)

θ (Yellow → No load , Blue → With load)



i_a (Yellow → No load , Blue → With load)

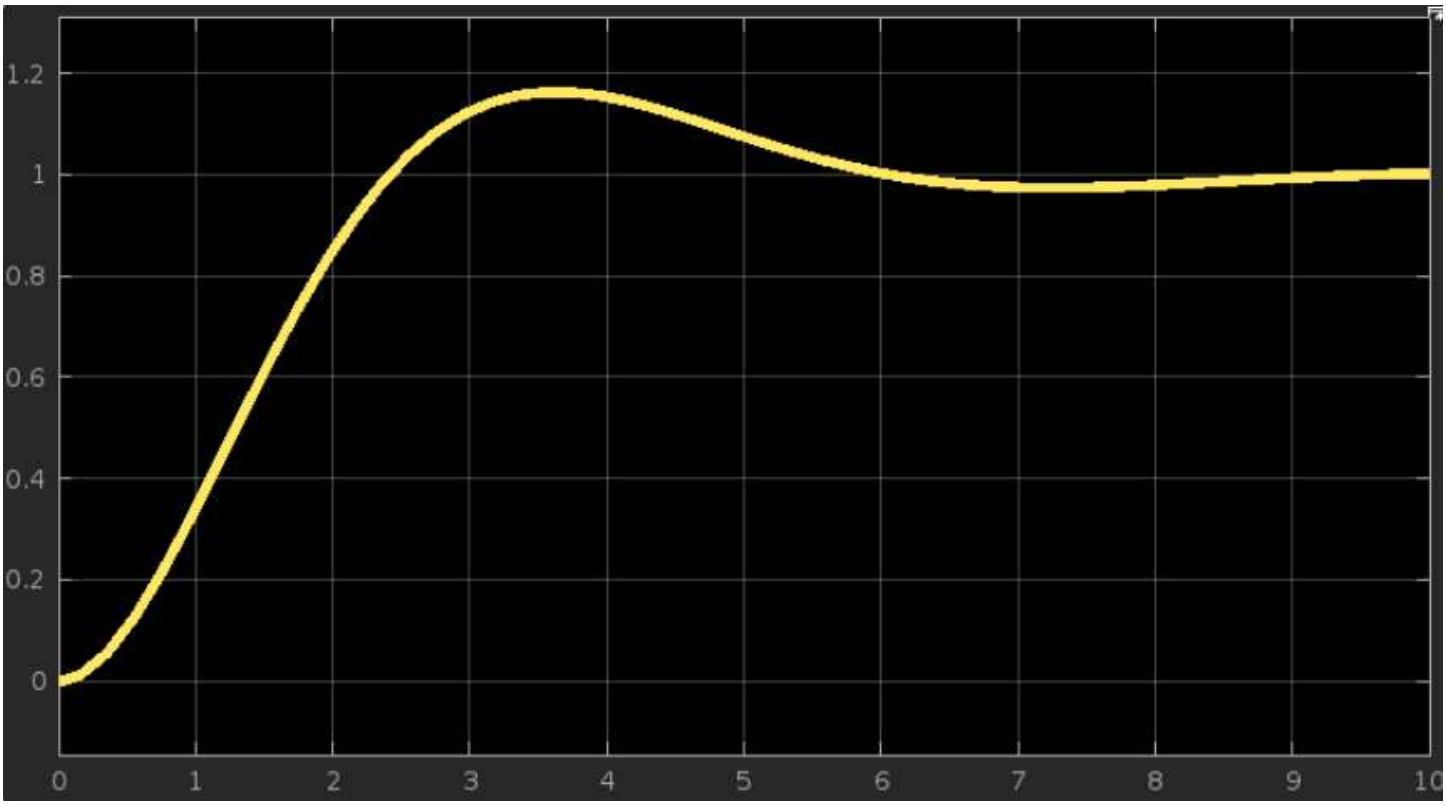


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Q4) The unit step response for a second order system shown below for $\omega_n = 1$ and $\zeta = 0.5$.



On the same figure plot the response for the following four systems:

1) $r(t) = 1 , \omega_n = 1 , \zeta = 1$
2) $r(t) = 1 , \omega_n = 1 , \zeta = 0$
3) $r(t) = 2 , \omega_n = 1 , \zeta = 0.5$
4) $r(t) = 1 , \omega_n = 2 , \zeta = 0.5$

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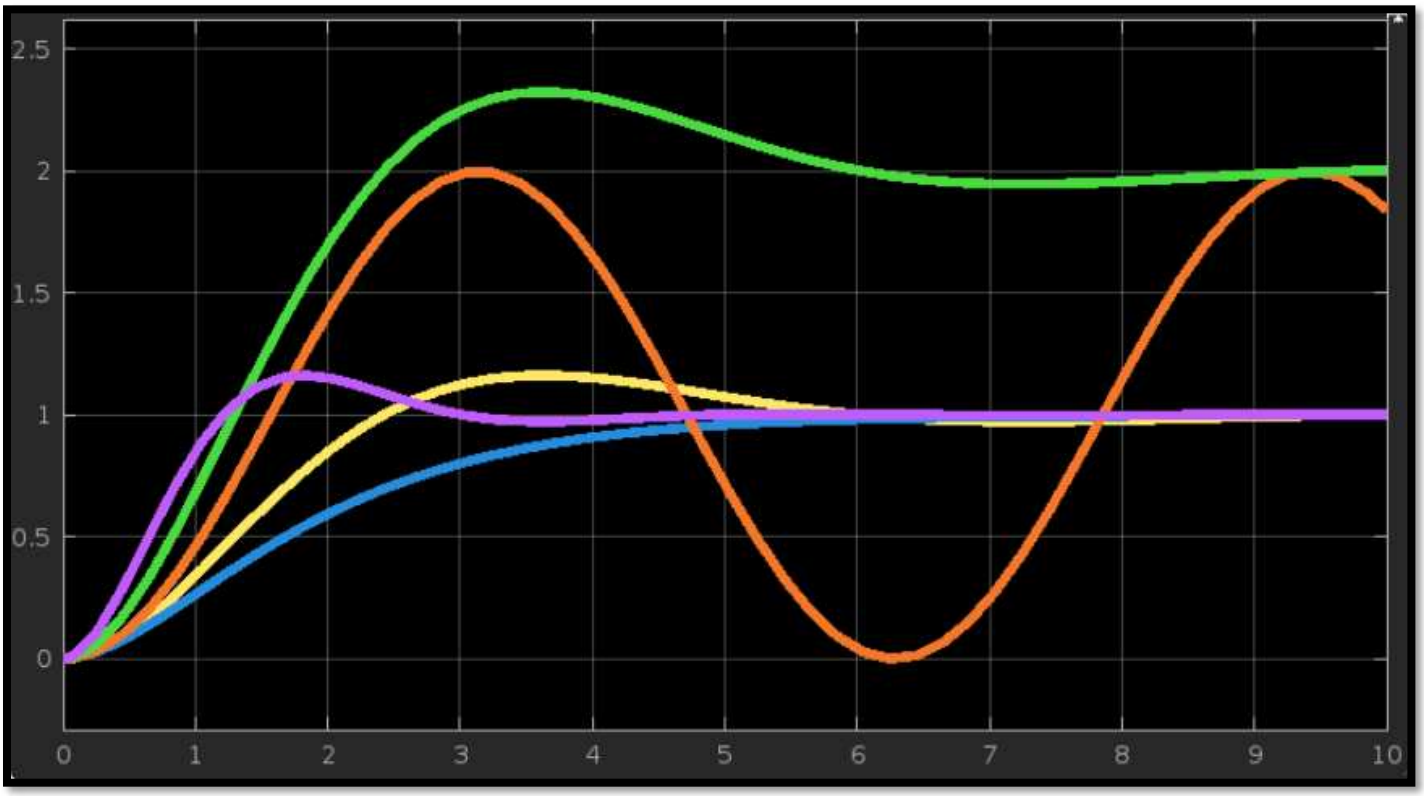
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Solution: (تأكدوا من الحل ولكن إن شاء الله إنه صح)

Original output $r(t) = 1 , \omega_n = 1 , \zeta = 0.5$ → Yellow

1) $r(t) = 1 , \omega_n = 1 , \zeta = 1$ → Blue
2) $r(t) = 1 , \omega_n = 1 , \zeta = 0$ → Orange
3) $r(t) = 2 , \omega_n = 1 , \zeta = 0.5$ → Green
4) $r(t) = 1 , \omega_n = 2 , \zeta = 0.5$ → Purple

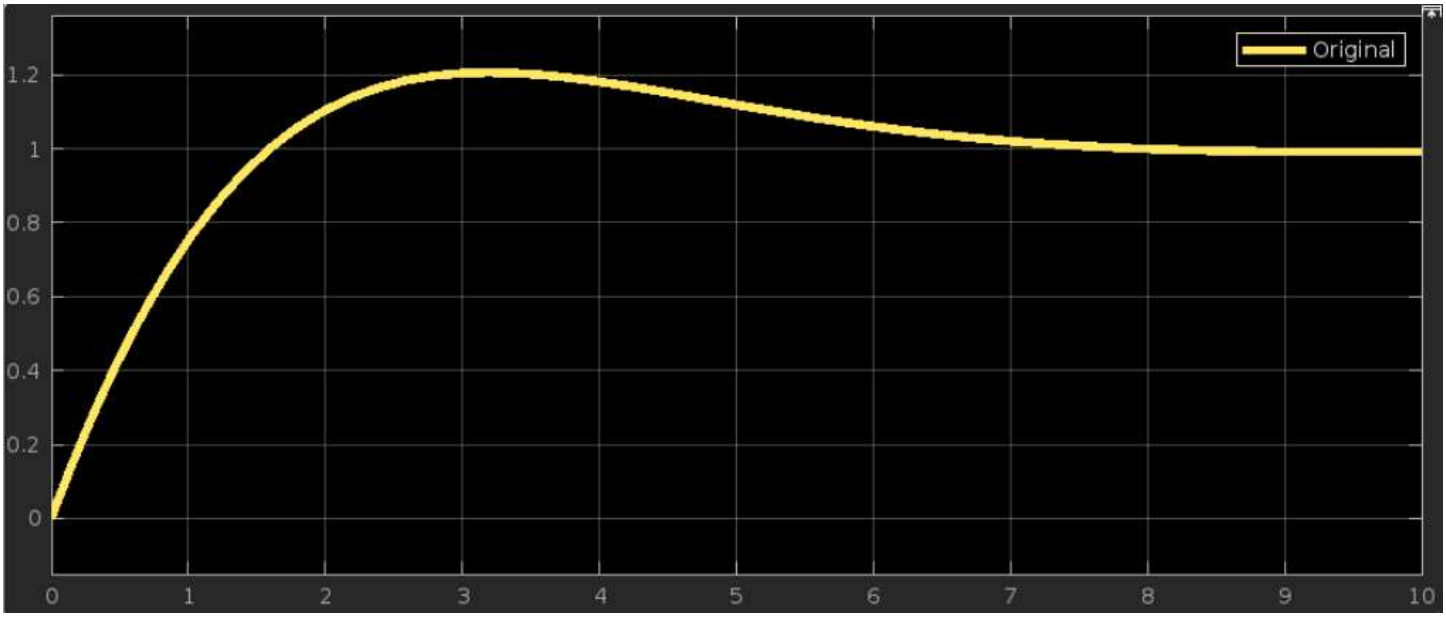


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Q5) PID Control output with unit step input shown in the figure below, discuss the following:



a) **Decreasing Kp**

b) **Increasing Ti**

c) **Decreasing Kd**

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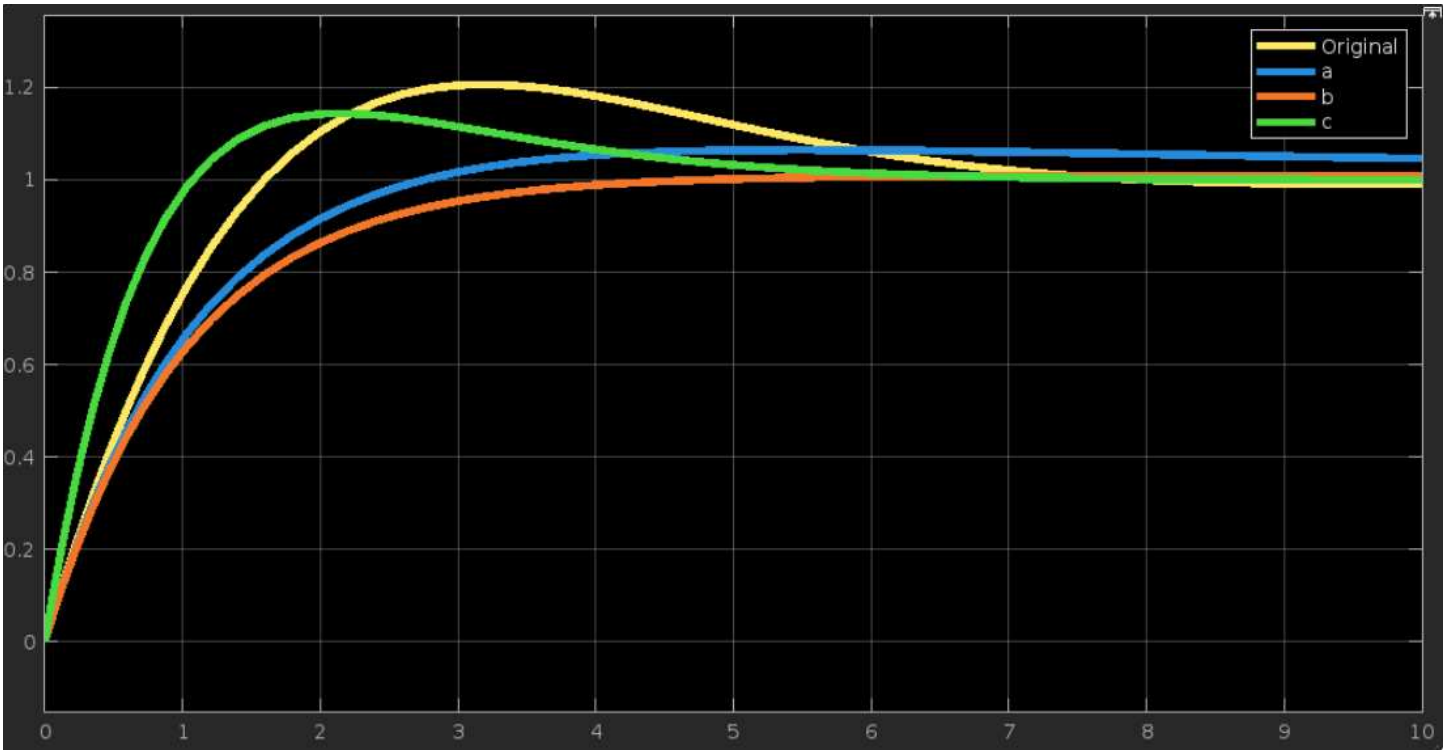
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Solution: (تأكدوا من الحل ولكن إن شاء الله إنه صح)

Original output → Yellow

a) Decresing Kp → Blue
b) Increasing Ti → Orange
c) Decresing Kd → Green



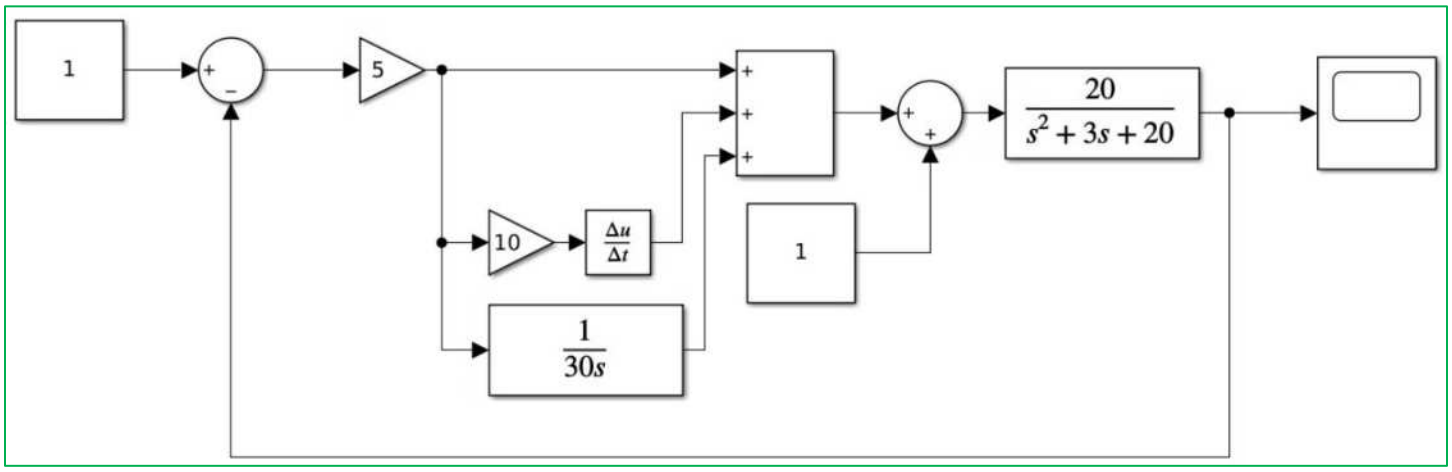
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Q6) Draw the unity feedback PID Controller with unit step input in MATLAB Simulink if:

$$G(S) = \frac{20}{S^2 + 3S + 20}$$

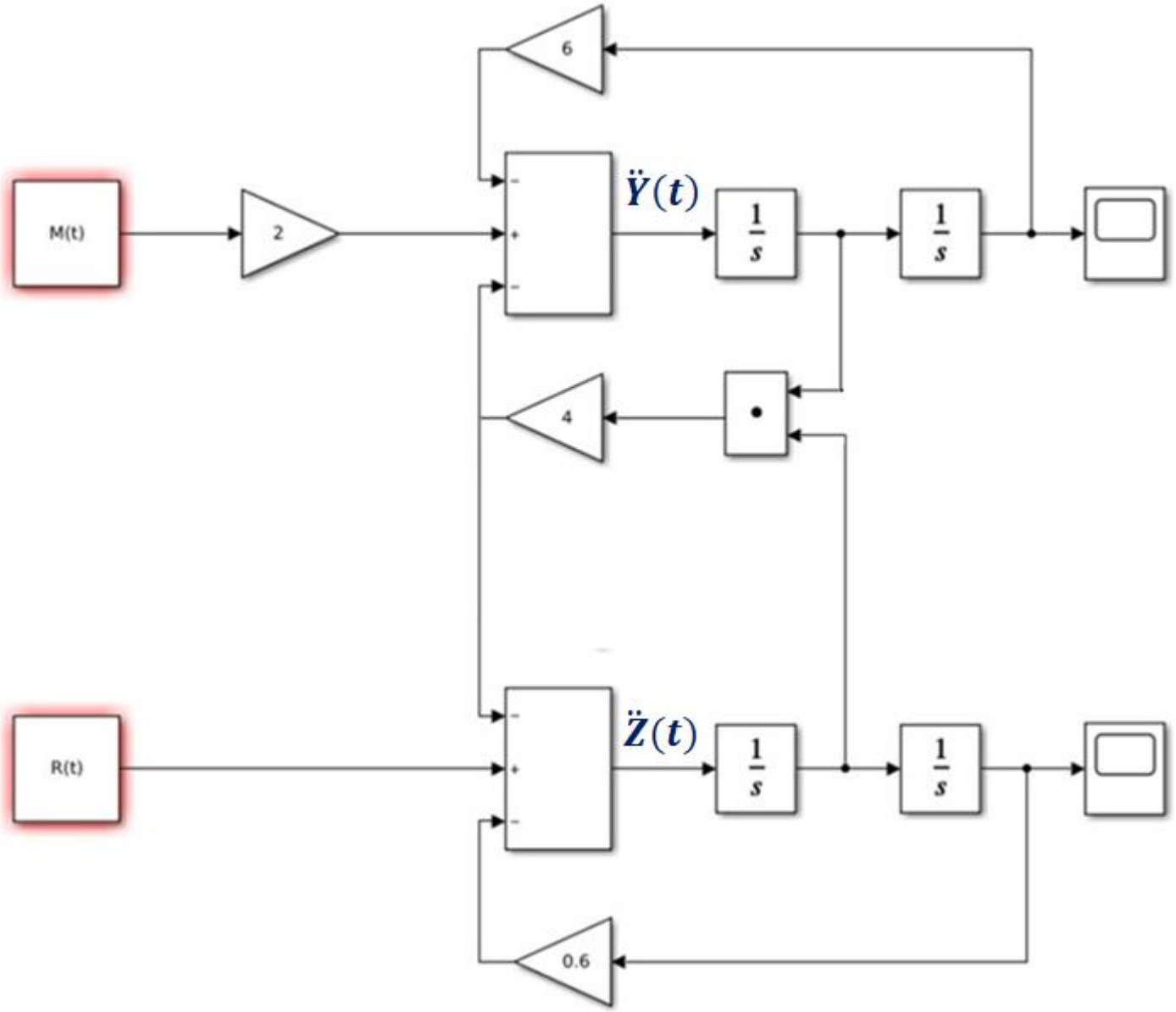
$$\frac{C(s)}{E(s)} = 5\left[1 + 10s + \frac{1}{30s}\right]$$

Solution:



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Q7) For the following systems represented by integration in Simulink, find the equation for $\ddot{Y}(t)$ and $\ddot{Z}(t)$:



Solution:

$$Y''(t) = 2M(t) - 4Y'(t)Z'(t) - 6Y(t)$$
$$Z''(t) = R(t) - 4Z'(t)Y'(t) - 0.6Z(t)$$

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Q8) Discuss the following:

a) Explain the effect of increased input in the first order system.

The final output in steady state increases, same response speed.

b) Explain the effect of increased T in the first order system.

Lower system response, long time to reach steady state.

c) Explain the effect of negative fixed feedback on time constant and gain.

Time constant → Decreased

Gain → Decreased

d) Explain the effect of negative flexible feedback on time constant and gain.

Time constant → Increased

Gain → No change

e) Explain the effect of positive fixed feedback on time constant and gain.

Time constant → Increased

Gain → Increased

f) Explain the effect of positive flexible feedback on time constant and gain.

Time constant → Decreased

Gain → No change

g) Using the I Controller for second order system.

The system unstable

h) When $\zeta = 0$.

Undamped

i) When increasing the Time Constant in First Order System.

The system response is slower, more time to reach steady state

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Q9) Choose the correct answer:

1. If we decrease load:

a. the speed increase and current decrease. b. the speed decrease and current increase.

c. the speed and current increase. d. the speed and current decrease.

2. If we increase load:

a. the position increase and current decrease. b. the position and current decrease.

c. the position decrease and current increase. d. the speed and current decrease.

3. If we integrated speed ω we get:

a. velocity. b. acceleration. c. inertia. **d. position.**

4. The transfer function $\frac{1}{Jm.s+Bm}$ represent:

a. amplifier. b. electrical motor. **c. gear.** d. load.

5. Decreasing the gain (Kp) in first order system with P-controller:

a. increase the steady state value. b. decrease the error.

c. increase settling time. d. decrease steady state value.

6. In Ziegler–Nichols second method, we keep increasing (Kp) from zero to (Kcr) value where the response becomes:

a. Overdamped. b. Underdamped. c. Unstable. **d. Undamped.**

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7. Decreasing the natural frequency ω_n in second order system:

a. increase peak time.

b. decrease settling time.

c. decrease error.

d. increase maximum overshoot.

8. Ziegler–Nichols PID Tuning methods aim to obtain:

a. 0% maximum overshoot.

b. 10% maximum overshoot.

c. 25% maximum overshoot.

d. 20% maximum overshoot.

9. If the damping ratio $\zeta = -0.2$, the system response is:

a. Undamped.

b. Critically damped.

c. Unstable.

d. Underdamped.

10. Increasing (Td) in second order system with PID - Controller:

a. decrease oscillations.

b. increase maximum overshoot.

c. unstable for all (Td) values.

d. decrease settling time.

11. The main use for I - Controller in PID - Controller is to:

a. decrease error.

b. decrease oscillations.

c. to lower the order of the system.

d. eliminating steady state error.

12. For a second order system, if the response was Overdamped, one of the following values is the damping ratio of the system:

a. $\zeta = 0$

b. $\zeta = 0.707$

c. $\zeta = 1.707$

d. $\zeta = 1$

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13. Ziegler – Nichols second method has been used on a system; the transfer function of the system was:

a. $G(s) = \frac{1}{3s+1}$

b. $G(s) = \frac{4}{s^2+6s+4}$

c. $G(s) = \frac{72}{s^3+20s^2+124s+24}$

d. $G(s) = \frac{1}{s}$

14. For second order system, if $\zeta = 0.5, t_{ss(2\%)} = 8s, \omega_n$ equals:

a. $\omega_n = 1$ b. $\omega_n = 0.75$ c. $\omega_n = 1.25$ d. $\omega_n = 0.39$

15. For $G(s) = \frac{1}{(s+1)(s+4)}$ we use Ziegler – Nichols':

a. first method.

b. second method.

c. we use another tuning method.

d. we don't use any method.

16. One of the following should be considered when using D controller:

a. Td value should be as high as possible.

b. Td value should be as low as possible.

c. using P controller with it.

d. Choosing Td value based on tuning method.

17. Given the damping ratio $\zeta = 0.3$, natural frequency $\omega_n = 3.1448$ the rise time t_r equals:

a. -23.13 s **b. 0.625 s** c. -25.82 s d. 23.13 s

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Q10) Based on the graph of an output signal of a system answer the following:

Rise Time =

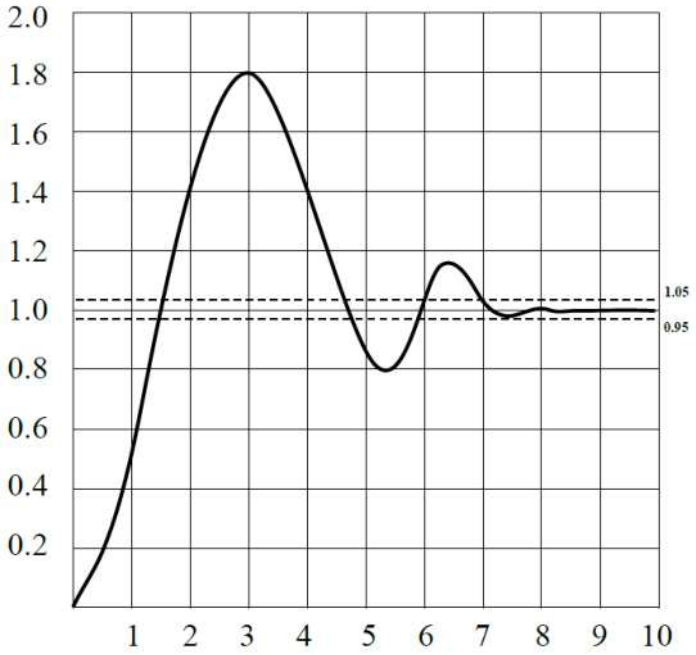
Peak Time =

Delay Time =

Maximum Overshot =

Type of response =

Steady-State Time (2% criterion) =



Solution:

Rise Time = 1.5s

Peak Time = 3s

Delay Time = 1s

Maximum Overshot = 0.8

Type of response = Underdamped

Steady-State Time (2% criterion) = 7s

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Q11) Answer the following questions:

a) **For Ziegler – Nichols second method, write down the steps of finding Kcr, Pcr values for PID controller parameters tuning.**

b) **For a Plant, we drew a tangent line at inflection point and the constants T and L respectively were T = 1.8, L = 0.3, using these values fill the table below:**

Type of controller	Kp	Ti	Td
P			
PI			
PID			

c) **When used Ziegler – Nichols second method to tune a PID controller, Pcr value was 2.5 sec, (Ti) value should be _____.**

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Solution:

a)

1. Setting Ti = ∞ and Td = 0
2. Increase Kp from 0 to Kcr until the system starts to oscillate steadily.
3. We find Kcr, now we need to find Pcr which means the time of signal takes for one full cycle of oscillation.
4. Using the following table for tuning Kp, Ti and Td:

Type of Controller	Kp	Ti	Td
P	0.5 × Kcr	∞	0
PI	0.45 × Kcr	Pcr / 1.2	0
PID	0.6 × Kcr	0.5 × Pcr	0.125Pcr

b) If we have T = 1.8 , L = 0.3 then we use it in First-Method (be aware):

Type of Controller	Kp	Ti	Td
P	6	∞	0
PI	5.4	1	0
PID	7.2	0.6	0.15

c) PID Controller in Second-Method Ti = 0.5 * Pcr → Ti = 1.25

تم بفضل الله ، اللهم ارحم عبدك (قاسم مفيد عمرو) واجعله من ورثة جنّة النعيم.

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